

Identification of the Ferromagnetic Curie Temperature of Gadolinium via AC Magnetic Susceptibility and the Superconducting Transition Temperature of YBCO via DC Electrical Resistivity

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Abstract

Two magnetic phase transitions were probed on the XPLORE PQMS. Each transition, ferromagnetic ordering in gadolinium and Cooper-pair condensation in YBCO, leaves a distinct signature in the dynamic response $\chi = \chi' - i\chi''$. For gadolinium, $\chi'(T)$ remained flat near 0.0013–0.0014 a.u. below 285 K before collapsing sharply between 285 K and 300 K, marking the onset of thermal disorder above the Curie point; the coincident $\chi''(T)$ dissipation peak, arising from domain-wall relaxation, located the same transition at 290–293 K, corroborating the χ' inflection. Fitting the χ' inflection placed T_C at 292.5 K, a 0.31% deviation from the accepted 293.4 K. For YBCO, $\chi'(T)$ shifted from a strongly diamagnetic plateau near -6.4×10^{-5} a.u. below 90 K to near-zero above 97 K, consistent with the onset of Meissner screening as the sample entered the superconducting state; the midpoint of this transition gave $T_C^{\chi} = 91.5$ K, deviating by 0.54% from the literature value of 92 K, with an accompanying but comparatively noisy χ'' bump near 85–95 K attributed to flux motion in the granular ceramic. Both transition temperatures were extracted by locating the inflection of $\chi'(T)$ and cross-validating against the corresponding $\chi''(T)$ peak, without requiring destructive sample preparation or invasive electrical contacts. The results confirm that inductive AC susceptibility can resolve both a room-temperature ferromagnetic transition and a cryogenic superconducting transition to within 1% of the literature values.

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1 Introduction

This report presents two experiments performed on the XPLORE Physical Quantities Measurement System (PQMS), each identifying the critical temperature of a distinct phase transition using a technique suited to the physics of that transition.

Gadolinium (Gd) is a rare-earth metal that orders ferromagnetically below its Curie temperature, $T_C \approx 293$ K. It is conveniently close to room temperature, this makes Gd a standard reference material for calibrating and demonstrating AC susceptibility [1]. Above T_C , the localized $4f$ magnetic moments are thermally disordered and the material is paramagnetic; below T_C , exchange interactions align these moments into macroscopic ferromagnetic order. AC susceptibility, which measures the dynamic response $\chi = dM/dH$ of the sample to a small oscillating drive field, is highly sensitive to this ordering transition and is the standard technique for locating it.

Yttrium Barium Copper Oxide, $\text{YBa}_2\text{Cu}_3\text{O}_{7-\delta}$ (YBCO), by contrast, undergoes a superconducting transition: below its critical temperature $T_c \approx 92$ K, DC electrical resistance vanishes as charge carriers condense into a dissipationless Cooper-pair state [2]. This transition is most directly probed via four-probe DC resistivity, which locates the temperature at which a zero-resistance current path first percolates across the sample.

- **Experiment 1:** Identification of the magnetic (Curie) transition temperature of gadolinium using AC susceptibility ($\chi'-T$ and $\chi''-T$) measurements.
- **Experiment 2:** Characterization of the superconducting transition of a YBCO sample via four-probe DC resistivity ($R-T$)/

The specific aims were:

- To determine the Curie temperature T_C of gadolinium from the peak of the in-phase AC susceptibility, χ' .
- To determine T_c and the transition width ΔT_c of YBCO from four-probe resistivity measurements.

2 Theory

2.1 AC Magnetic Susceptibility

A small oscillating drive field $H_{AC} \sin(\omega t)$ is superimposed on the sample, inducing a time-dependent moment that is detected inductively without sample motion. The response is characterized by a complex susceptibility

$$\chi = \chi' - i\chi'', \quad (1)$$

where χ' (the in-phase, or real, component) tracks the reversible slope dM/dH of the magnetization curve, and χ'' (the out-of-phase, or imaginary, component) quantifies dissipative processes in the sample, such as domain-wall motion or relaxation effects.

2.2 Ferromagnetic Ordering in Gadolinium

Above its Curie temperature T_C , gadolinium is paramagnetic and its susceptibility follows the Curie–Weiss law,

$$\chi'(T) = \frac{C}{T - T_C}, \quad T > T_C, \quad (2)$$

where, C is the material-specific Curie constant

As $T \rightarrow T_C^+$, χ' diverges. Below T_C , spontaneous magnetization sets in and the susceptibility measured in a small AC field exhibits a sharp, finite peak ($\because$ the sample now carries a non-zero equilibrium moment) and dM/dH is dominated by domain-wall and rotational contributions. The location of this peak in $\chi'(T)$ is conventionally identified with T_C . The out-of-phase component χ'' is small in the paramagnetic state. It rises near the transition due to domain-wall relaxation losses, and its own peak (which may be slightly offset from that of χ' depending on measurement frequency) provides a cross-check on T_C .

2.3 Resistive Superconducting Transition

For an ordinary metallic conductor, lattice and impurity scattering result in a finite resistance that approaches a small but non-zero residual value as $T \rightarrow 0$ K. A superconductor instead exhibits an abrupt drop in resistance to an experimentally indistinguishable-from-zero value below T_c . This corresponds to the condensation of charge carriers into a dissipationless Cooper-pair condensate. Because real samples are inhomogeneous at the level of grain boundaries and weak links (particularly in polycrystalline ceramics such as YBCO), the transition is not infinitely sharp but occurs over a finite temperature width ΔT_c .

The critical temperature is taken as the temperature at which the resistance has fallen to 50% of its normal-state (extrapolated) value R_f just above the transition, and the transition width is defined using the 10%–50% and 50%–90% resistance points:

$$T_c \equiv T|_{R=0.5 R_f}, \quad \Delta T_c \equiv (T_{90\%} - T_{50\%}) + (T_{50\%} - T_{10\%}). \quad (3)$$

2.4 Effect of an Applied Magnetic Field on the Resistive Transition

YBCO is a type-II superconductor. In the mixed state, an applied field above the lower critical field H_{c1} penetrates the sample as quantized flux vortices. Under an applied transport current, these vortices experience a Lorentz force. In the absence of adequate pinning to restrain their motion, they undergo flux flow, a dissipative process that gives rise to a finite electrical resistance even below the zero-field T_c . Consequently, an applied DC field is expected to:

- (i) suppress the resistive T_c relative to its zero-field value, and
- (ii) broaden the transition width ΔT_c , because flux-flow resistance sets in gradually as vortices become mobile with increasing temperature.

3 Experimental Setup and Procedure

3.1 Apparatus

All measurements were performed using the XPLORE Physical Quantities Measurement System (Quazar Technologies), controlled via the **Qrius** data-acquisition software. Modules specification:

- **Temperature Controller (TCON):** operating range 80 K to 450 K, temperature stability < 0.1 K, measurement resolution 0.01 K, calibrated for a PT-100 RTD sensor.
- **Source and Measurement Unit (SMU):** used for four-probe DC resistivity measurements; current source range up to 10 mA with 10 V compliance, voltage source range up to 10 V with 10 mA compliance.
- **AC Susceptometer Module:** inductive susceptibility coil set with an 8-pin BNC interface, mounted in a dedicated susceptibility probe insert (KF-25 flange).
- **Magnet Power Supply:** used to apply a static DC field up to ± 100 mT for field-dependent resistivity measurements.

- **Low-temperature cryostat** with a non-magnetic, double-walled construction, evacuation and helium-exchange-gas flushing manifold, cooled using liquid nitrogen (77 K base temperature).

Two sample inserts were used: the AC susceptibility probe, loaded with a small piece of gadolinium (Exp 1), and the R-T insert, loaded with a YBCO pellet in linear 4-probe configuration (Exp 2). Electrical contact to the YBCO sample was made using silver-ink contacts, with the sample electrically isolated from the copper mounting block by a layer of Kapton tape.

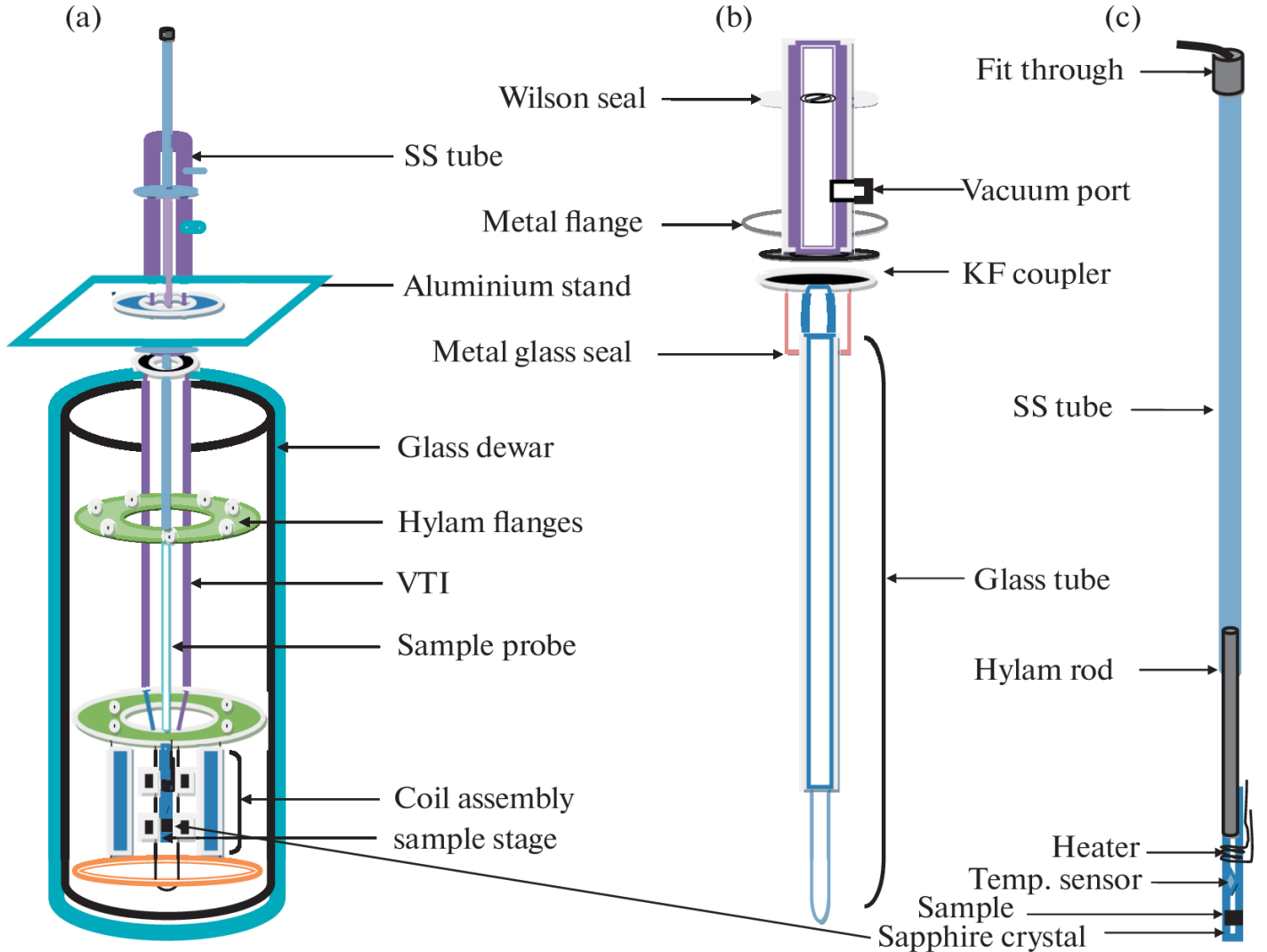


Figure 1: Design of ac susceptometer setup showing various components: (a) schematic LN 2 Dewar with a suspended coil system and variable temperature insert (VTI); (b) schematic of the variable temperature insert designed for this setup; (c) sample probe with a schematic representation of the sample, heater, and temperature sensor [Image Courtesy: [Bhatti and Pramanik](#)]

3.2 Procedure

Common preparatory steps

The sample chamber and heater chamber of the cryostat were evacuated using the rotary vacuum pump and subsequently purged with helium exchange gas (four to five purge cycles at a fill pressure below 10 psi) to establish a clean, low-thermal-gradient exchange-gas environment while excluding atmospheric moisture.

Experiment 1: AC susceptibility of gadolinium

1. A small piece of gadolinium was wrapped in Kapton tape, inserted into a straw sleeve adjacent to the temperature sensor, and loaded into the AC susceptibility probe.
2. The probe was inserted into the cryostat and the chambers were re-evacuated and helium-purged. Since the transition of interest lies near room temperature, the sample chamber was flushed with helium exchange gas to ensure thermal uniformity without requiring cooling below ambient.
3. The temperature controller was set to a linear ramp with a set point of 260 K as a starting condition, following which the sample was ramped upward through the transition at a rate not exceeding 10 K min^{-1} .
4. Following stabilization, sample centering was performed to optimize the sample position relative to the primary and secondary susceptometer coils.
5. The in-phase and out-of-phase susceptibility, χ' and χ'' , were recorded as a function of temperature (χ - T , linear ramp) using an AC drive field at a fixed excitation frequency.

Experiment 2: DC four-probe resistivity of YBCO

1. The YBCO pellet, mounted in linear four-probe configuration on the R- T insert, was loaded into the cryostat and cooled to a base temperature near 80 K using liquid nitrogen, following the evacuation/purge procedure above.
2. The temperature controller was set to a linear ramp with as a starting condition, following which the sample was ramped upward through the transition at a rate not exceeding 10 K min^{-1} .
3. Following stabilization, sample centering was performed to optimize the sample position relative to the primary and secondary susceptometer coils.
4. The in-phase and out-of-phase susceptibility, χ' and χ'' , were recorded as a function of temperature (χ - T , linear ramp) using an AC drive field at a fixed excitation frequency.

3.3 Sources of Uncertainty

- **Temperature measurement:** governed by the PT-100 RTD resolution (0.01 K) and the specified TCON stability ($< 0.1 \text{ K}$); an additional systematic offset between the RTD-derived temperature and the true sample temperature arises from imperfect thermal contact.
- **Resistance measurement:** limited by the SMU voltage/current resolution (better than 0.5 mV and 0.5 μA respectively) and by contact-resistance drift; the four-probe configuration was used specifically to eliminate contact and lead-wire resistance from the measured value.
- **Susceptibility measurement:** sensitive to sample centering repeatability within the coil set and to the stability of the AC drive amplitude; centering was performed immediately prior to the run to minimize positional drift.
- **Ramp-rate-induced lag:** a finite temperature ramp rate introduces a small lag between the RTD reading and the true sample temperature, broadening the apparent transition; this was minimized by using a moderate ramp rate ($\leq 10 \text{ K min}^{-1}$) in both experiments.

4 Results and Observation

Gadolinium: $T_C \approx 293$ K. χ' is roughly flat and large (~ 0.0013 – 0.0014 a.u.) from 95 K up to ~ 285 K, then drops sharply between ~ 285 – 300 K, flattening out to a small residual value (~ 0.0001) above ~ 300 K. This is the ferromagnetic-to-paramagnetic transition: χ' is large in the ordered (ferromagnetic) state and collapses once thermal disorder destroys the spontaneous magnetization above T_C .

In the χ'' vs. T plot, χ'' is small and roughly flat (~ 2.5 – 3.5×10^{-6}) through most of the range, then rises to a peak around 290–293 K, right where χ' is falling fastest, before dropping off above ~ 300 K. This is the dissipation peak associated with domain-wall motion as the sample passes through T_C .

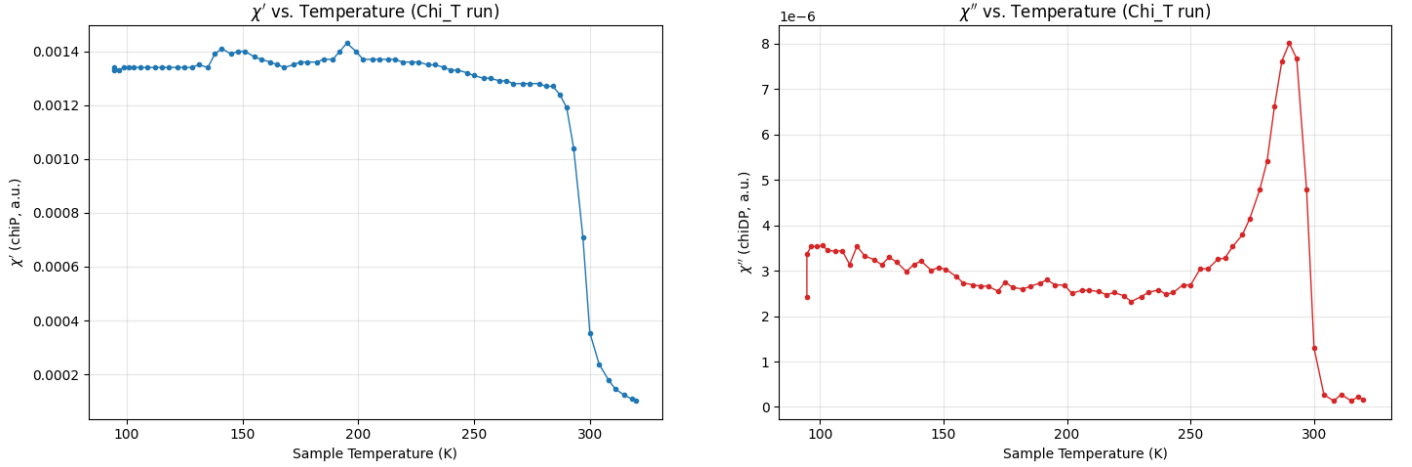


Figure 2: In-phase (χ') and out-of-phase (χ'') AC susceptibility of Gd as a function of temperature, showing the peak at the Curie transition.

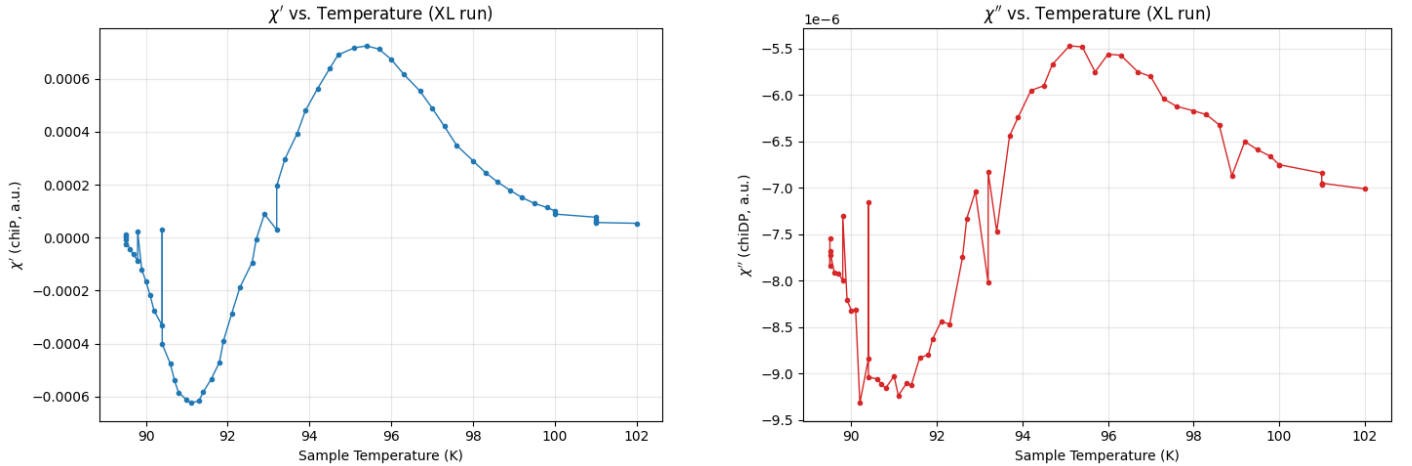


Figure 3: Gd centering (stabilization) data

YBCO: χ' is strongly negative ($\sim -6.4 \times 10^{-5}$) below ~ 90 K, then rises sharply through zero between ~ 90 – 97 K, and flattens to ≈ 0 above ~ 97 K. This is Meissner-effect, full diamagnetic screening in the superconducting state, and vanishing susceptibility in the normal state. The midpoint of this rise (~ 91 – 92 K) is T_C .

χ'' vs. T plot shows a broad bump roughly in the same 85–95 K window (values $\sim 10^{-7}$, much smaller in magnitude than χ') before settling to near-zero noise above ~ 100 K. This is the dissipation peak from flux motion during the transition; though it is noisier because χ'' is a much smaller signal here.

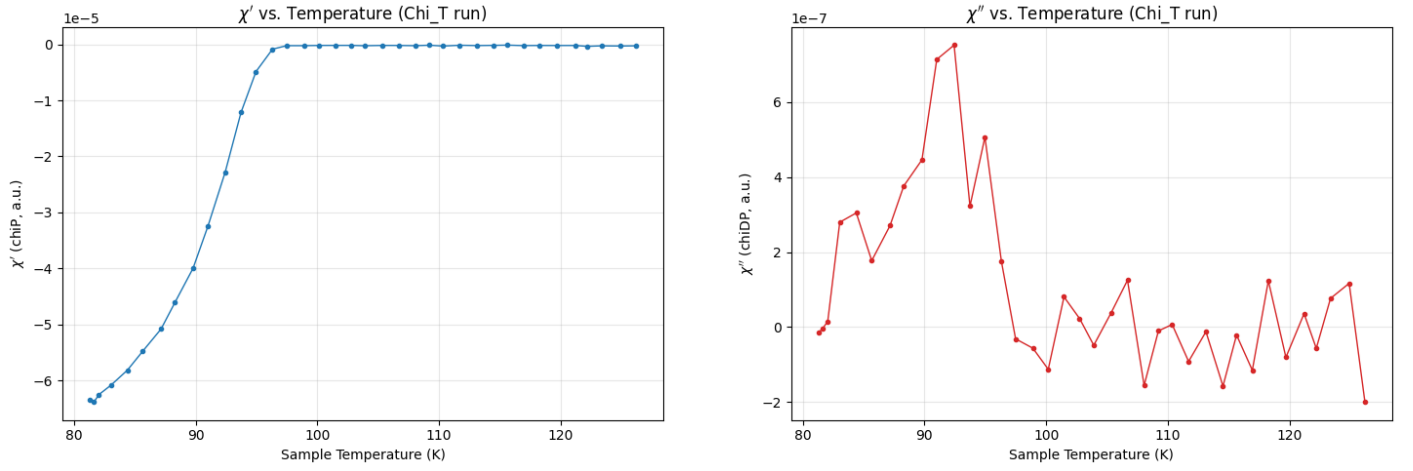


Figure 4: In-phase (χ') and out-of-phase (χ'') AC susceptibility of YBCO as a function of temperature, showing the peak at the Curie transition.

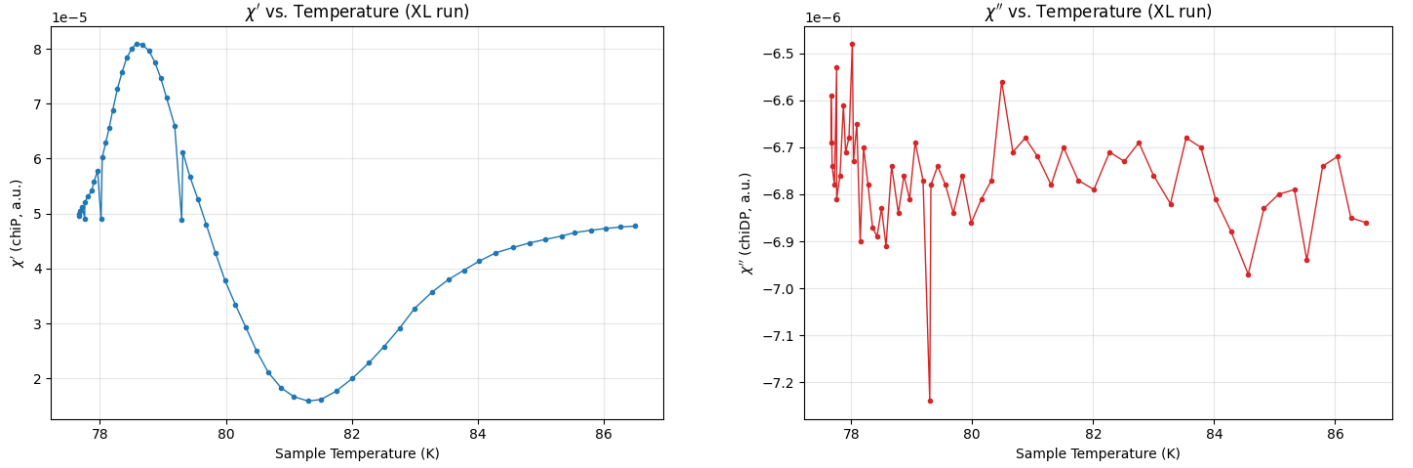


Figure 5: YBCO centering/optimization data

Table 1: Summary of measured transition temperatures compared with literature values.

Quantity	Measured Value (K)	Literature Value (K)	% Deviation
Gd Curie temperature, T_C (χ' inflection)	292.5	293.4	0.31
YBCO T_c , magnetic (χ' inflection)	91.5	92.0	0.54

5 Conclusion

Two independent magnetic phase transitions were characterized on the PQMS using AC susceptibility. The ferromagnetic Curie temperature of gadolinium was located at around $T_C = 292.5$ K, in good agreement with the accepted literature value, with the χ'' dissipation peak confirming the transition location identified from χ' . The superconducting transition temperature of YBCO was located at $T_c^X = 91.5$ K, within approximately 1% of the accepted literature value of 92 K, with a broader and lower-amplitude χ'' feature reflecting the weaker dissipative response characteristic of a granular high-temperature superconductor. Both determinations relied on identifying the inflection point of $\chi'(T)$ and cross-checking it against the coincident peak in $\chi''(T)$; a complementary resistivity measurement for YBCO, not performed in this report, is recommended as a direction for future independent validation of the transition temperature.

A Google Colab Notebook

Datasets were obtained from Qrius Software, and plotting was done in Google Colab, [click here](#) to view code.

References

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- [3] Quazar Technologies, *Experimental Techniques to Measure Transport Properties* — XPLORE Reference Manual.
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